INCH-POUND
MIL-PRF-55371B(CR)
3 JULY 1997
SUPERSEDING
MIL-T-55371A(CR)
24 May 1991

PERFORMANCE SPECIFICATION
TAPE, RADIO FREQUENCY MEASURING
(P/O ANTENNA GROUP, AN/GRA-50 ( ))

This performance specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers the design and construction requirements for a special purpose measuring tape which is a part of the Antenna Group, AN/GRA-50( ).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2). Handbooks are for guidance only and therefore are not mandatory.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-454 - Standard general requirements for electronic equipment.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

Drawings

SC-A-46439

(Copies of documents, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Where specified in the contract (see 6.2), the contractor shall furnish first article units for inspection (see 4.2).

3.2 Materials. The contractor shall select the materials, but the materials and parts used shall be fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but are not mandatory. Selection criteria of the class, grade or type part shall be that the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.3 Description. This item is a special version of a measuring tape (General Use) (see 6.2.i) modified to accurately calibrate military antennas in the field. The tape described herein consists of a case, holding strap, reel, handle, and a ribbon. The ribbon is graduated with

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linear dimensions (feet) on one side and corresponding frequency markings (megahertz) on the other side.

3.4 Construction. The Radio Frequency Measuring Tape shall be constructed in accordance with the requirements of this specification and shall be capable of passing verification prior to delivery to the government (see 4. VERIFICATION).

3.5 Performance characteristics.

3.5.1 Graduations and Figures. Tapes shall be graduated for a length of 156 feet. Graduations and Figures (numbers and letters) shall be permanent and resistant to wear and abrasion resulting during normal and abnormal use in a wide range of environmental conditions. Graduation lines shall be uniform and readily distinguishable in their widths. These lines shall be free from discontinuities and ragged edges. There shall be no noticeable variation in the widths of like numbers and letters. Each graduation shall be accurately located. Tapes shall be graduated for reading left-hand to right-hand.

3.5.2 Finish. All exposed metal parts shall be constructed of corrosion-resistant material or shall be plated with a corrosion resistant plating such as zinc, chromium or nickel. All exposed metal parts shall be manufactured free of defects, sharp edges, burrs, or any other physical imperfections which could impair manual handling, or injure the operator (see 3.7).

3.5.3 Case. The case shall house the ribbon and provide for the handling of the tape. The case shall be consructed of corrosion-resisting metal or a metal with corrosion finish, and shall be not less than .025 inches thick. Choice of construction is at the option of the contractor, but the opening in the case intended for the ribbon shall be provided with a durable eyelet having rollers to act as bearing surfaces on top and bottom sides facing the ribbon. The bearing surface rollers shall facilitate the deployment of the ribbon when used. The case shall be capable of withstanding the shock; drop and bounce tests of 4.6.4 et el.

3.5.4 Case Covering. The vnyl cover shall be vnyl coated cloth which is both durable and water resistant. A stainlesss steel or plated steel furling ring shall be used to cover and securely hold the edges of the vnyl coated cloth to the case. The vinyl coated shall be mildew resistant. The vinyl case cover shall be permanently secured to the case with a water adhesive.

3.5.5 Holding strap. A 150 pound tensile strength holding strap, measuring 18 x 1/2 x 1/16 inches, shall be provided across the back of the case. The strap shall be wrapped around two rollers 5 and 3/8 inches apart. Each roller shall be free to rotate about a pin shaft which shall be mechanically fastened (e.g. riveted) at both ends to a bracket. Each bracket shall be mounted to the case via three mechanical fasteners. The strap shall be retained by two sliding loops. Associated hardware may be of any material which is strong, durable and corrosion resistant. The strap shall be resistant to ultraviolet deterioration, fungus growth, abrasion, and moisture, and pliable in temperatures as low as -50 degrees F.

3.5.6 Identification marking. Each tape and ribbon shall be plainly and permanently marked with the manufacturer’s name or with a trademark of such known character that the

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source of manufacture may be determined. Each tape shall also be legibly marked on the outside of the case in a plain and permanent manner with the length of the tape.

3.5.7 Ribbon assembly. The ribbon shall be constructed of synthetic fiber (e.g. unwoven fiberglass, polyvinyl coated or similar material). All the threads used in the construction of the ribbon shall be fungus resistant. The ribbon shall be between 1/2 and 7/8 inches wide, and the woven tapes shall have a selvage (woven edge) located at each edge. The ribbon shall be easily replaced in the tape, when required.

3.5.7.1 Ribbon strength. The strength of the ribbon shall be sufficient to withstand a 80 pound load applied to each end. This required strength shall be demonstrated via testing in compliance with 4.5.4.

3.5.7.2 Ribbon abrasion resistance. The ribbon shall be constructed to resist abrasion due to normal to heavy use in extreme environmental conditions. The normal process of deploying and reeling up the ribbon with the handle will produce some apparent normal wear over years of use and shall be considered in the construction of the ribbon and the quality of the graduations and markings placed on the ribbon. When the ribbon is deployed with any over a portion of its entire length of 156 feet, the surface of the ribbon may be subject to abnormal abrasion when it makes contact with the ground. The ribbon portion may suffer most significant abrasion when its either dragged over the ground during use or while being reeled. The aforementioned abrasion situations shall be provided for in the design and construction of the ribbon and its graduations. A documented demonstration of abrasion resistance shall be performed by the contractor on selected samples of graduated ribbons (in accordance with 4.5.5). The useful life of the ribbon shall be ten (10) years.

3.5.7.3 Ribbon finger ring. A corrosion resistant ring shall be attached to the outer end of the ribbon large enough for a finger hold and strong enough to prevent breakage. The ring shall be fastened to the ribbon in a manner which permanently affixes the ring to the ribbon. The ring shall be the same width as the ribbon. The finger ring shall withstand the following test: Each sample Tape ribbon shall be supported at 4-1/2 inches from the tip of the finger ring and a 40 pound test load applied across the finger ring to 75 percent of its width, for not less than five minutes. The ribbon shall not break or be pulled out of the clip fastened to the finger ring, and the finger ring shall not break or deform when tested in accordance with 4.5.7.

3.5.7.4 Reinforcement. Reinforcement shall be provided on the outer end of the ribbon for not less than 4 inches by a strip of leather, plastic or other abrasion resistant material of the same width as the ribbon. The reinforcement shall be fungus and mildew resistant. The reinforcement shall pass around the ring and under the clip.

3.5.8 Graduations. One side of the ribbon shall have graduations marked in feet, tenths and half-tenths of a foot. The graduation lines and tenth of a foot indication numerals shall be in black, and the foot indication numerals shall be in red. The outside end of the ring shall be the point of origin or “zero” point.

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3.5.8.1 First foot of ribbon. Each tenth graduation shall be numbered to indicate tenths of a foot. The tenth of ten graduations shall be numbered to indicate feet. The first end marking up to 2/10 of a foot may be obliterated by end fitting.

3.5.8.2 Second and each successive foot of ribbon. The second to ninth graduations of each successive foot shall be numbered to indicate feet and tenths of a foot. Tenth graduations shall be numbered to indicate feet.

3.5.9 Frequency markings. The reverse face of the ribbon shall be graduated to frequency in megahertz. The frequency markings shall be abbreviated “MHz” and shall be as listed in Table I. The location of the markings shall correspond to the listed linear dimension on the other side of the ribbon. Each numerical graduation shall include the abbreviation “MHz”. All characters shall be black and not less than 1/4 inch high.

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TABLE I. Frequency markings.

Frequency (MHz)	Length (Ft)	Frequency (MHz)	Length (Ft)	Frequency (MHz)	Length (Ft)
1.5	156.0	2.3	103.73	5.4	43.33
1.52	153.94	2.35	99.57	5.6	41.78
1.54	151.94	2.4	97.50	5.8	40.35
1.56	150.0	2.45	95.51	6.0	39.0
1.58	148.10	2.5	93.60	6.2	37.74
1.60	146.25	2.55	91.76	6.4	36.56
1.62	144.44	2.6	90.00	6.6	35.45
1.64	142.68	2.65	88.30	6.8	34.41
1.66	140.96	2.7	86.66	7.0	33.42
1.68	139.28	2.75	85.09	7.5	31.20
1.70	137.64	2.8	83.57	8.0	29.25
1.72	136.04	2.85	82.10	8.5	27.52
1.74	134.48	2.9	80.68	9.0	26.0
1.76	132.95	2.95	79.32	9.5	24.63
1.78	131.46	3.0	78.0	100	23.40
1.80	130.0	3.1	75.48	10.5	22.29
1.82	128.57	3.2	73.12	11.0	21.27
1.84	127.17	3.3	70.9	12.0	19.51
1.86	125.80	3.4	68.82	13.0	18.0
1.88	124.46	3.5	66.85	14.0	16.71
1.90	123.15	3.6	65.0	15.0	15.60
1.92	121.87	3.7	63.24	16.0	14.62
1.94	120.61	3.8	61.57	17.0	13.76
1.96	119.38	3.9	60.0	18.0	13.0
1.98	118.18	4.0	58.50	19.0	12.31
2.0	117.0	4.2	55.71	20.0	11.7
2.05	114.14	4.4	53.18		
2.1	111.42	4.6	50.86		
2.15	108.83	4.8	48.75		
2.2	106.36	5.0	46.80		
2.25	104.0	5.2	45.0		

3.5.10 Reel. The reel shall hold the ribbon in the case and be capable of unwinding and winding the ribbon during use. Deployment of the ribbon shall be via the ribbon finger ring (3.5.7.3). Winding of the ribbon shall be via the handle (3.5.11). The reel shall be of durable and rugged construction and be free to rotate without restriction or binding. Construction material may be of any non-corrosive material that can withstand the high loads and resultant high stresses which result from usage defined in 3.5.7.1 and 3.5.7.2. The reel shall have a holding strength of 40 pounds applied through a sample tape ribbon supported by the reel with the ribbon connecting strap fully pulled out for a period of 5 minutes. During this proof test, the ribbon shall not become loose or become detached from the reel. The reel shall show no signs of binding, slipping, or failure.

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3.5.11 Handle. The handle shall be suitable for winding the ribbon onto the reel (see 3.5.10). The handle shall neatly fold against the reel for storage and shall have a handcrank of at least 1 inch long. The handle shall be of rugged metal construction which shall not rust, tarnish, corrode, or become pitted when subjected to the environment. The winding of the ribbon shall be smooth and unrestricted when using the handle.

3.5.12 Accuracy. The error in the length of the ribbon on a horizontal surface with a tension of 2 pounds at a temperature of 68°F or 20°C shall not exceed 2 inches per 100 feet when tested in accordance with 4.5.2.

3.5.13 Error due to change in humidity. The error shall not exceed 0.25% when tested in accordance with 4.5.3.

3.5.14 Operation. The ribbon shall be capable of being extended and read up to the graduated length of 156 feet and then be retracted completely within the case using the handle (see 3.5.11) and shall pass the test of 4.5.1.

3.5.15 Useful life. The useful service life of the Tape shall be ten (10) years.

3.6 Environmental conditions.

3.6.1 Low temperature. The tape shall show no evidence of damage, deterioration of the winding mechanism or degradation of performance when subjected to operation or storage at temperatures as low as -80°F for a period of 24 hours.

3.6.2 High temperature. The tape shall show no evidence of damage, deterioration of the winding mechanism or degradation of performance when subjected to operation or storage at temperatures as high as 125°F for a period of 24 hours.

3.6.3 Humidity (moisture resistance). The tape shall show no evidence of damage, deterioration of the winding mechanism or degradation of performance when subjected to relative humidities of 95% @125°F for a period of 48 hours.

3.6.4 Shock; drop and bounce.

3.6.4.1 Shock; drop and bounce. The tape shall show no evidence of damage, deterioration of the winding mechanism, or degradation of performance when subjected to the test of 4.6.4.1.

3.6.4.2 Bounce. The tape shall show no window of damage in the test of 4.6.4.2.

3.7 Workmanship. All tapes shall be manufactured in such a manner as to be uniform in quality and shall be free of defects that will affect tape life, serviceability, interchangeability, or appearance. MIL-HDBK-454, requirement 9 is recommended as guidance only, not mandatory for satisfactory performance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.4).

4.2 First article. First article inspection shall be performed on a specified number of Tapes when a first article sample is required (see 3.1).

4.2.1 First article units. The contractor shall furnish a specified number first article units of the complete Radio Frequency Measuring Tape as required by the contract or purchase order.

4.2.2 First article inspection. The first article inspection shall consist of the visual examination as specified in 4.3 and the inspections specified in Table II which shall be performed on each unit in the order specified therein.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Shock; drop	3.6.4.1	4.6.4.1
Bounce	3.6.4.2	4.6.4.2
Visual and mechanical (workmanship)	3.7	4.7
Operation	3.5.14	4.5.1
Accuracy	3.5.12	4.5.2
Finger ring	3.5.7.3	4.5.7
Holding strength at reel	3.5.10	4.5.6
Temperature	3.6.1/3.6.2	4.6.1/4.6.2
Humidity (moisture-resistance)	3.6.3	4.6.3
Error due to change in humidity	3.5.13	4.5.3
Tensile strength of ribbon	3.5.7.1	4.5.4
Abrasion test	3.5.7.2	4.5.5

4.2.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.3 Visual inspection. Each tape shall be visually examined for workmanship per 3.7 or any other nonconformance in design, finish, coating, construction, weight, and marking requirements for which specific tests are unnecessary. Specific defects to be visually inspected are cited in 4.7.

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4.4 Conformance inspection. The contractor shall perform the inspections specified in 4.4.1 through 4.4.5. This does not relieve the contractor of the responsibility for performing any additional inspections necessary to control the quality of the product and to assure compliance with all specification requirements.

4.4.1 Group A inspection. Units presented for inspection shall be inspected for conformance to the criteria specified in Table III. Lots in which samples exhibit any failures shall be screened for that failure prior to being subjected to Group B inspection.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Visual and Mechanical (workmanship)	3.7	4.7
Case	3.5.3	4.7
Holding strap	3.5.5	4.7
Ribbon	3.5.7	4.7
Handle	3.5.11	4.7
Operation	3.5	4.5.1
Case, Vinyl cover	3.5.4	4.7

4.4.1.1 Order of inspection within Group A. Group A inspection shall be performed in any order satisfactory to the Government.

4.4.2 Group B inspection. This inspection shall consist of the inspections listed in Table IV and shall be performed on lots that have passed Group A inspection. Lots in which samples exhibit any failures shall be screened for that failure prior to the units within the lot being subjected to Group C inspection.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test paragraph
Accuracy	3.5.12	4.5.2
Finger Ring	3.5.7.3	4.5.7
Holding strength at reel	3.5.10	4.5.6

4.4.2.1 Order of inspection within Group B. Group B inspection shall be performed in any order.

4.4.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and B inspections, and shall consist of the inspection specified in Table V. Sampling shall be in accordance with 4.4.3.1.

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TABLE V. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Shock; drop	3.6.4.1	4.6.4.1
Bounce	3.6.4.2	4.6.4.2
Temperature	3.6.1/3.6.1/2	4.6.1/4.6.2
Humidity (moisture -resistance)	3.6.3	4.6.3
Error due to change in humidity	3.5.13	4.5.3
Tensile strength of ribbon	3.5.7.1	4.5.4
Abrasion test	3.5.7.2	4.5.5

4.4.3.1 Sampling for Group C inspection. Sampling shall follow the procedures indicated in 6.2.j.

4.4.3.2 Order of inspection within Group C. Group C inspection shall be performed on each unit in the order shown in Table V.

4.4.3.3 Group C failures. Group C failures shall be reported and corrective action taken in accordance with the requirements specified in the contract or purchase order.

4.4.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the samples units are resubjected to and pass Group A and B inspections.

4.5 Methods of inspection.

4.5.1 Operation test. The ribbon shall be withdrawn for its entire length and retracted using the handle. There shall be no evidence of restriction or binding when performing this test. (See 3.5.14).

4.5.2 Accuracy. The ribbon shall be extended to its full length and tested for accuracy of graduations and frequency markings in accordance with 3.5.12.

4.5.3 Error due to change in humidity. A specimen 20 inches in length shall be removed from each sample tape under test and conditioned for 48 hours in circulating air of 30 ± 2 percent relative humidity at a temperature of 70° to 80°F. The specimen shall be hung in a vertical position under no other load than its own weight during the conditioning period. At the end of the 48 hours, a 10 pound load shall be hung on the lower end of each specimen and after a period of 10 minutes, the distance between the gage marks measured. The load shall then be removed and the specimen conditioned for 48 hours in circulating air 95 ± 2 percent relative humidity and at a temperature of 70° to 80°F. At the end of 48 hours of conditioning, the distance between the measurements taken shall be considered as the change in linear dimensions due to humidity and all samples shall conform to 3.5.13.

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4.5.4 Tensile strength of ribbon. The tensile strength of the ribbon shall be determined for compliance with the tensile strength requirements of 3.5.7.1 as follows: Four specimens, each 0.6 foot long shall be cut from the ribbon. Each specimen shall be creased transversely in two places 0.1 foot apart near the center of the specimen. The creases shall be made by folding the tape 180° with firm finger pressure on the crease. One fold shall be made back to back and the other to front, the front being the face with the numerals. The folds shall then be straightened with finger pressure and the specimen placed in the tensile tester with the section containing the straightened folds between the jaws.

4.5.5 Abrasion test. Three specimens 21 inches long shall be cut from the ribbon of each sample tape. One of these tapes shall contain and have this test performed on a frequency marking. The other two specimens shall contain and have this test performed on graduations. The other two specimens shall contain and have this test performed on graduations. The specimens are to be tested as follows: Immerse the brush bristles in water at 77° to 86°F for 30 minutes to a depth of one-half inch, and then rub Bon Ami cake soap into the brush for about 30 seconds, working the soap well into the bristles by using the corners and edges of the cake. The brush should be turned on its back during this operation so that no more than a minimum amount of water is lost. Mount the three specimens on the glass washability panel with waterproof tape, insert the brush in its holder and start the motor. The brush shall travel at a rate of 45 ± 5 cycles (90 separate strokes ° 10 strokes) per minute. After each 150 cycles, wash and resoap the brush. After a total of 1000 cycles, the 150 cycles, wash and resoap the brush. After a total of 1000 cycles, the outline of any number, letter or graduation in the center eight inches of scrub stroke of each specimen shall be completely discernible. (See 3.5.7.2).

4.5.6 Holding strength at reel. Each sample tape shall be supported at the reel and with the ribbon and connecting strip fully pulled out, a 25 pound load shall be applied to the ribbon for not less than five minutes. The ribbon shall not be loosened from the reel. (See 3.5.10).

4.5.7 Finger ring test. Each sample tape shall be supported at 4-1/2 inches from the tip of the finger ring and a 40 pound test load applied across the finger ring to 75 percent of its width, for not less than five minutes. The tape shall not break or be pulled out of the chip which fastens the finger ring, and the finger ring shall not break or deform. (See 3.5.7.3).

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4.6 Environmental tests. Environmental testing shall be performed in accordance with the requirements of this specification, except when a contractor's own test plan showing equivalent methods has been approved by the Government contracting office. When the contractor's test plan deviates from the test procedures prescribed, the contractor shall be required to show where his own tests test methods for verifying the specified performance are equivalent. Also, measurements taken by the contractor during verification testing apparatus, e.g. environmental test chamber, vibration test table, pendulum shock device, or similar test equipment for which the technical capabilities to properly simulate extreme climatic or transportation conditions are scientifically documented. The methods for testing to verify compliance with all stated performance characteristics selected by the contractor shall be equivalent to the recommended procedures found herein.

4.6.1 Low temperature. Verify low temperature performance using the requirement of 3.6.1 via performing the tests of the sort prescribed in 6.3. Failure to meet the requirement shall constitute failure of the sample.

4.6.2 High temperature. Measure compliance with the high temperature requirement of 3.6.2 via performing the tests of the sort prescribed in 6.3. Failure to meet the requirement shall constitute failure of the sample.

4.6.3 Humidity (moisture-resistance). Assure compliance with the humidity requirement of 3.6.3 via performing the tests of the sort prescribed in 6.3. Failure to meet the requirement shall constitute failure of the sample.

4.6.4 Shock; drop and bounce tests.

4.6.4.1 Shock; drop. The tape shall be subjected to the following 14 drops, each from a height of 48-inches. The floor or barrier receiving the impact shall be 2-inch fir wood backed by concrete or a rigid frame.

a. Drop on each face (2 drops).

b. Drop flat on edge, 4 points 90-degrees apart, starting at ribbon exit area (4 drops).

c. Drop on corners to either side of the edge drops of b above (8 drops).

After the test, the tape shall meet the requirements of 3.6.4

4.6.4.2 Bounce. The Tape shall be tested on the package tester, Type 1000-SC as made by the L.A.B. Corporation, Skaneateles, New York, or equal. accessories shall be selected from those listed on SC-A-46439. The test shall be performed as follows:

a. Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.
b. Place the tape on the bed of the package tester. Limit the lateral motion, by wooden

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fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to safeguard personnel, provided that the fore-and-aft motion of the adapter plate against the backstop is not restrained.

c. Operate the package tester, shafts in phase, for a total of 3 hours at 283 ±1 rpm.

After the test the tape shall meet the requirements of 3.6.4.

4.7 Visual and mechanical inspection. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship and marking. Defects are listed in Table VI (see 3.7).

TABLE VI. Classification of visual and mechanical defects.

Classification	Defects
All Major	1. Incomplete, component parts missing.
	2. Ribbon material not as specified; evidence of torn or frayed edges, selvage missing; ribbon broken, cracks.
	3. Ring not securely attached to ribbon, not reinforced as specified.
	4. Graduations and figures incomplete, illegible, on wrong side of tape, not as specified.
	5. Case thickness less than the minimum specified; case not covered as specified. 6. Holding strap not as specified.
	7. Vinyl cover incomplete, evidence of uncoated spots, furling ring nonconforming.
	8. Ribbon width nonconforming.
	9. Opening for ribbon not provided with an eyelet as specified; rollers do not revolve, rollers not smooth, evidence of burrs.
	10. Ribbon not attached to reel as specified.
	11. Reel winding device not as specified.
	12. Handling fails to hold against reel, crank length nonconforming; pin or button fails to release handle from locked position.
	13. Evidence of hazardous sharp edges or burrs on exposed metal parts.
Minor	1. Finish of exposed metal parts not as specified.
	2. Marking, manufacturer's name or trademark missing, illegible or not permanent.

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4.8 Group C inspection. Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action (see 4.4.3.3).

4.9 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this inspection and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

4.10 Sampling. Sampling shall follow the procedures cited in 6.2.j.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's system Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The tape is part of the Antenna Group AN/GRA-50 ( ) and is used to measure antenna element length.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. First article inspection: Two sample units of the Radio Frequency Measuring Tape are generally required (see 4.2).

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature required.

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g. Quantity of tools and running spare parts required.

h. Tape, Measuring (General Use), GGG-T-106, Type I, Class B and Case V.

i Sampling for verification inspection should conform to ANSI/ASQC Z1.4, sampling procedures and tables for inspecting attributes.

6.3 Environmental inspection procedures. The test methods and procedures of MIL-STD-810 listed below are recommended, but not mandatory for determining compliance with the environmental requirements of this specification. The contractor may use any test method that he can verify as providing comparable stress levels on the material to be tested following Government approval of the test plan.

Inspection	Specification Paragraph	MIL-STD-810 Method	MIL-STD-810 Procedure	Remarks
Low temperatures	3.6.1	502	I	Use Steps 1,2,6, 7 and 8. Storage temp. shall be -80°F and duration of Step 2 is 24 hours.
High temperature	3.6.2	501	I	Use Steps 1,2,6,7 and 8. During of Step 2 shall be 24 hours.
Humidity	3.6.3	507	II	Use Steps 1,5,6, and 9.

6.4 Definitions.

a. Damage. Breakage, loosening, shifting, evidence of rust or corrosion, the failure of any part, connection, hardware, surface finish; or the failure of any component to perform its intended function.

b. Degradation. Lessened or reduced levels of performance produced as a direct result of testing procedure or measurement process performed as part of verification. Degradation will be considered a failure to meet specified performance levels.

6.5 Subject term (key word) listing.

Frequency markings

Graduation and figures
Ribbon assembly

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6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CUSTODIAN:
Army-CR

PREPARING ACTIVITY:
Army-CR

(Project 5210-0291)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-55371B	**2. DOCUMENT DATE (YYMMDD)** 97/07/03

3. DOCUMENT TITLE TAPE, RADIO FREQUENCY MEASURING (P/0 ANTENNA GROUP, AN/GRA-50()

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME ROGER LOUISAIRE US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-2735 (2) AUTOVON 992-2735
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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